FREEDOM INTERNATIONAL SCHOOL

WORKSHEET
PHYSICS
CLASS XI
THERMAL PROPERTIES OF MATTER

1. Pendulum clocks generally run fast in winter and slow in summer. Why?
2. What kind of thermal conductivity and specific heat requirements would you specify for cooking utensils?
3. A metal ball is heated through a certain temperature. Out of mass, radius, surface area and volume, which will undergo largest percentage increase and which one the least?
4. Can we boil water inside an artificial satellite of the earth?
5. A refrigerator converts 50g of water at 15ºC into ice at 0ºC in one hour. Calculate the quantity of heat removed per minute. Take specific heat of water = 1cal $g^{-1}{}^{o}C^{-1}$ and latent heat of ice = 80 cal g^{-1}.
6. Which object will cool faster when kept in open air, the one at 300^0C or the one at 100^0C? Why?
7. A hole is drilled in a copper sheet. The diameter of the hole is 4.25 cm at $27^0 C$.What is the change in the diameter of the hole when the sheet is heated to $227^0 C$? Coefficient of linear expansion of copper is $1.70 \times 10^{-5}\ {}^0C^{-1}$.
8. A copper block of mass 2.5 kg is heated in a furnace to a temperature of 500^0C and then placed on a large ice block. What is the maximum amount of ice that can melt?
   (Specific heat of copper = 0.39 $Jg^{-1}{}^{o}C^{-1}$ and heat of fusion of water = 335 Jg^{-1})
9. Two stars radiate maximum energy at wavelengths 3.6×10^{-7}m and 4.8×10^{-7}m respectively. What is the ratio of their temperatures?
10. A large steel wheel is to be fitted on to a shaft of the same material. At 27ºC, the outer diameter of the shaft is 8.70 cm and the diameter of the central hole in the wheel is 8.69 cm. The shaft is cooled using 'dry ice'. At what temperature of the shaft does the wheel slip on the shaft? Assume coefficient of linear expansion of the steel to the constant over the required temperature range.
    ($\alpha_{steel} = 1.20\times10^{-5}K^{-1}$)